FYS5419/9419, lecture May 14, 2025

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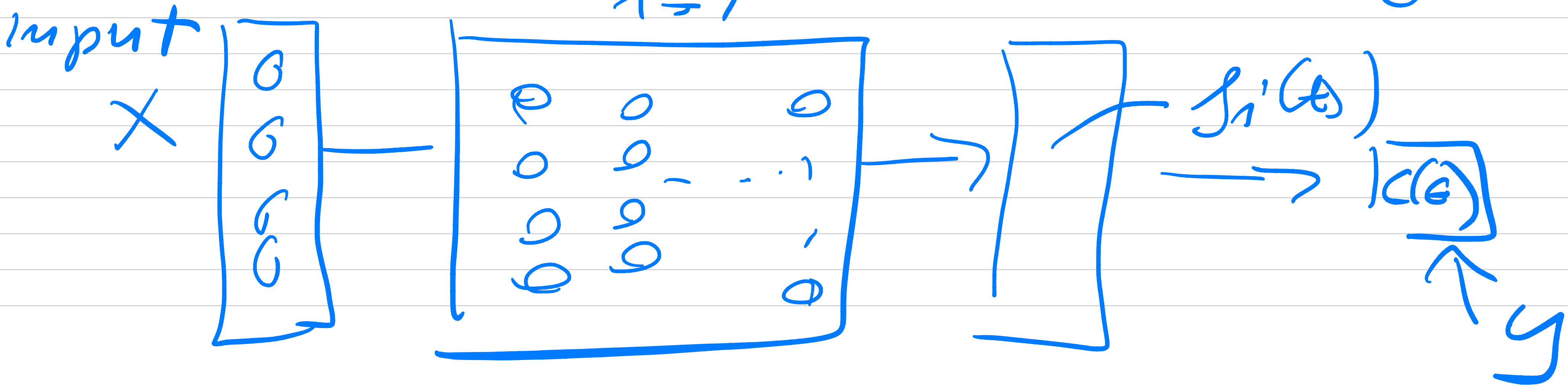
Training set X, y $(y_i = \{0,1\})$

model $f_i(\theta)$

$$C(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - f_i(\theta)\right)^2$$

$$\nabla_\theta C(\theta) = -\sum_{i=1}^{n}\left(y_i - f_i(\theta)\right)\nabla_\theta f_i(\theta)$$



input data is encoded into a state $|\psi_i\rangle$

$$f_i(\theta) = \langle \psi_i | U^\dagger(\theta) \hat{O} U(\theta) | \psi_i \rangle$$

The angles θ Layers

$$\theta = \begin{bmatrix} \theta_{11} & \theta_{12} & \cdots & \theta_{1\ell} \\ \theta_{21} & & & \\ \vdots & & & \\ \theta_{n1} & \cdots & \cdots & \theta_{n\ell} \end{bmatrix}$$

$$\theta \in \mathbb{C}^{n \times \ell}$$

gate operation with parameter θ_{ij}

$$G(\theta_{ij}) \quad \{ R_{xyz}(\theta_{ij}\,\sigma_{xyz}) \}$$

$$U(\theta) = V\,G(\theta_{ij})\,W$$

gate sequence

W precedes $G(\theta_{ij})$

V comes after.

$$\hat{Q} = V^{+}\hat{O}V$$

$$|\phi_i'\rangle = W|\psi_i'\rangle$$

$$\frac{\partial f_i'(\epsilon)}{\partial \epsilon_{i'j'}} = \frac{\partial}{\partial \epsilon_{i'j'}}\Big(\langle \phi_i' | G(\epsilon_j)^{+}\hat{Q}$$

$$\times\, G(\epsilon_{i'j'})|\phi_i\rangle\Big)$$

$$= \langle \phi_i | \left(\frac{\partial}{\partial \epsilon_{ij}} G^{+}(\epsilon_{ij}) \right) \hat{Q}\, G(\epsilon_{ij}) | \phi_i \rangle$$

$$+ \langle \phi_i | G^{+}(\epsilon_{ij})\, \hat{Q} \left(\frac{\partial}{\partial \epsilon_{ij}} G(\epsilon_{ij}) \right) | \phi_i \rangle$$

$$B = G(\epsilon_{ij}) \qquad C = \frac{\partial G(\epsilon_{ij})}{\partial \epsilon_{ij}}$$

$$\langle \phi_i | C^{+} \hat{Q} B | \phi_i \rangle + \langle \phi_i | B^{+} Q C | \phi_i \rangle$$

$$= \frac{1}{2} \left(\langle \phi_i | (B+C)^{+} \hat{Q} (B+C) | \phi_i \rangle - \right.$$

$$\langle \phi_i \,|\, (B-C)^{+} \, \hat{Q} \, (B-C) \,|\, \phi_i \rangle \Big)$$

parameter-shift rule,

$$G(\theta_{ij}) = \exp(-i\,\theta_{ij}\,\Gamma)$$

$$\Gamma = \sigma_x,\ \sigma_y,\ \sigma_z,\ \mathbb{1}$$

$$\frac{\partial f_1(\theta)}{\partial \theta_{ij}} = \frac{2}{2}\Big[\langle \phi_i | \left(\mathbb{1} - \frac{i}{2}\Gamma\right)^{+} \hat{Q}$$

$$\times \left(\mathbb{1} - \frac{i}{2}\Gamma\right) |\phi_i\rangle$$

$$- \langle \phi_i | \left(\mathbb{1} + \frac{i}{2} \Pi \right)^{+} \hat{Q} \left(\mathbb{1} + \frac{i}{2} \Pi \right) | \phi_i \rangle \Big]$$

$$s = \frac{\pi}{42} \qquad 2 = \pm 1/2$$

$$s = \pm \pi/2$$

$$\frac{\partial f_i}{\partial \theta_{ij}} = 2 \Big(\langle \phi_i | G^{+} (\theta_{ij} + s) \hat{Q}$$

$$\times G(\theta_{ij} + s) | \phi_i \rangle$$

$$- \langle \phi_i | G^{+} (\theta_{ij} - s) \hat{Q} \, G(\theta_{ij} - s) | \phi_i \rangle \Big)$$

$$\frac{\partial f_i(\theta)}{\partial \theta_{ij}'} = \frac{1}{2}\Big(\langle \phi_i | G^{+}\left(\theta_{ij}' + \frac{\pi}{2}\right)$$

$$\times \hat{Q}\, G\left(\theta_{ij}' + \frac{\pi}{2}\right) | \phi_i' \rangle$$

$$- \langle \phi_i | G^{+}\left(\theta_{ij} - \frac{\pi}{2}\right) \hat{Q}\, G\left(\theta_{ij}' - \frac{\pi}{2}\right)$$

$$\times | \phi_i' \rangle \Big)$$

$$P_\theta(x_i) = \frac{f(x_i;\theta)}{Z(\theta)}$$

normalization

$$P_\theta(x) = \prod_{x_i \in D} P_\theta(x_i)$$

$$P_\theta(x_i, h_i)$$

$$P_\theta(x_i) = \sum_{h_i \in H} P_\theta(x_i, h_i)$$

$$Z(\theta) = \sum_{\substack{x_i \in D \\ h_j \in H}} e^{f(x_i, h_j; \theta)}$$

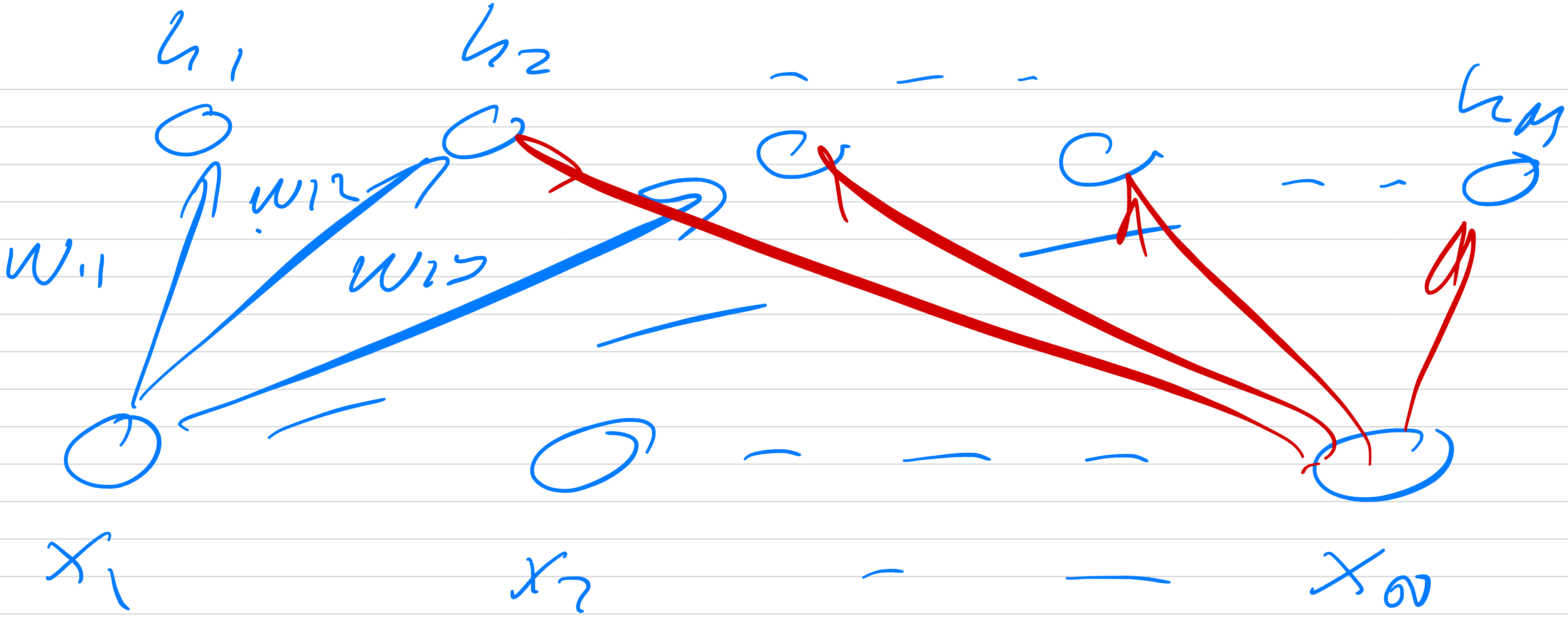
$$f(\vec{x}_i, \vec{h}_j; \theta) = \sum_{i=1}^{N} a_i x_i + \sum_{j=1}^{M} b_j h_j + \sum_{ij}^{NM} w_{ij} x_i h_j$$

$$\theta = \{\vec{a}, \vec{b}, W\}$$

Binary - Binary

$$x_i = \{-1, +1\}$$

$$h_j = \{-1, +1\}$$



$$\hat{\theta} = \arg\max_{\theta} P_{\theta}(x)$$

$$\hat{\theta} = \arg\max_{\theta} \log P_{\theta}(x)$$